



Tool wear prediction in milling operation

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WHAT SHOULD BE FOUND

1. Material Removal Rate (MRR)

- Measures how much material is removed per unit of time

2. Cutting Time

- Time required to complete the milling operation.

Problems faced

- **1. Reduced Dimensional Accuracy**

Worn tools can result in inaccurate cuts, leading to out-of-tolerance parts.

This may necessitate rework or scrapping of the component.

- **2. Poor Surface Finish**

Tool wear often causes increased roughness on the machined surface.

Chipping or rounding of cutting edges can leave irregular marks.

- **3. Increased Cutting Forces**

As the tool wears, cutting edges lose sharpness, increasing resistance during cutting.

This puts more load on the machine and tool, accelerating further wear or potential breakage.

- **4. Higher Heat Generation**

Worn tools are less efficient at cutting, leading to increased friction and temperature.

Excessive heat can damage both the tool and the workpiece (e.g., thermal deformation, microstructural changes).

- **5. Tool Breakage**

Progressive wear, especially if undetected, can lead to sudden and catastrophic tool failure.

This not only damages the tool but may also harm the workpiece and machine.



Dataset features

- **Spindle speed**
 - **Feed rate**
 - **Tool material**
 - **Coolant usage**
 - **Workpiece hardness**
 - **Depth of cut (mm)**
 - **Cut time**
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Regression algorithm

- Linear regression

Linear Regression models the relationship between a **dependent variable (target)** and one or more **independent variables (features)** by fitting a straight line.

- Lasso regression

Lasso Regression adds an **L1 penalty** (absolute values of coefficients). It can **shrink some coefficients to zero**, effectively performing **feature selection**.

- Ridge regression

Ridge Regression adds an **L2 penalty** (squared magnitude of coefficients) to the loss function. It helps reduce overfitting by shrinking coefficients but **never makes them zero**.

- Elastic net regression

ElasticNet combines both **L1 and L2 penalties**, offering a balance between Ridge and Lasso. It is effective when you have **many correlated features** and want both **regularization and feature selection**.



BEST ALGORITHM

- Linear Regression R^2 Score:

1.0 MSE: 8.78664129632872e-31

The **best algorithm** for predicting **Tool Wear** depends on your linear regression algorithm. However, based on your described features — both **numerical and categorical**, with likely **nonlinear interactions**

BENEFITS FOR PREDICTING

